Capstone Project Concept Note and Implementation Plan

Project Title: Syria Wheat Climate Intelligence (SWCI): AI-Based Climate Risk Prediction for Wheat Yield in Syria

Team Members:
Haya Alwahb
Diana Salman
Dana Azizieh

CONCEPT NOTE

1. Project Overview

Wheat production in Syria is vital for national food security but is highly vulnerable to climate variability such as drought and irregular rainfall. Syria Wheat Climate Intelligence (SWCI) is a machine learning proposal designed to evaluate climate risks and predict wheat yield. The project directly aligns with SDG 2 (Zero Hunger) and SDG 13 (Climate Action).

2. Objectives

Develop an ML regression pipeline to estimate annual wheat yield based on climate indicators.
Establish a historical-average baseline to evaluate performance using MAE and RMSE metrics under time-aware validation.
Build a Streamlit prototype dashboard for decision-makers to explore climate scenarios.

3. Background

Syrian wheat production relies heavily on rainfall, making it susceptible to climate shocks. Machine learning provides non-linear analysis between climatic variables (rainfall and temperature peaks) and crop yields, offering data-driven tools for agricultural adaptation.

4. Methodology

The workflow includes data cleaning, growing-season feature engineering (e.g., Nov–Apr rainfall), baseline development, and Random Forest Regression modeling. Validation uses a time-based train-test split to prevent data leakage. Performance is evaluated via MAE and RMSE metrics.

5. Architecture Design Diagram

[Data Sources (CCKP & FAOSTAT)] -> [Preprocessing & Feature Engineering]
-> [Random Forest Model vs Baseline] -> [Streamlit UI Dashboard]

Data Sources: Ingests climate and agricultural records.
Preprocessing & Features: Cleans annual data and extracts growing-season metrics.
Modeling: Trains Random Forest Regressor against a historical baseline.
User Interface: Streamlit dashboard rendering climate scenarios.

6. Data Sources

Data is sourced from the World Bank CCKP (historical temperature and precipitation) and FAOSTAT (wheat yield and harvested area). Preprocessing aggregates metrics into a unified annual country-level dataset.

7. Literature Review

Studies confirm ensemble models like Random Forest outperform simple linear regressions in modeling non-linear climate impacts on crop yields. This project adapts these methods to Syria's specific growing season.

IMPLEMENTATION PLAN

1. Technology Stack

Language & Libraries: Python, Pandas, NumPy, Scikit-Learn, Matplotlib

UI & Control: Streamlit, Git & GitHub

2. Timeline

Stage 1: Data Collection | Download and merge CCKP & FAOSTAT datasets | Phase 1

Stage 2: EDA & Features | Extract growing-season metrics and analyze correlation | Phase 2

Stage 3: Modeling | Train Baseline & Random Forest; compute MAE & RMSE | Phase 3

Stage 4: UI & Delivery | Build Streamlit dashboard and complete GitHub documentation | Phase 4

3. Milestones

M1: Integrated climate and wheat yield dataset ready.

M2: Random Forest model outperforming the baseline.

M3: Functional interactive Streamlit prototype deployed.

4. Challenges and Mitigations

Data Gaps: Handled via linear interpolation and historical checks.

Overfitting: Mitigated by constraining tree depth in Random Forest.

5. Ethical Considerations

Uses public aggregated data to ensure privacy. Model predictions include confidence bounds to avoid misleading decisions.

6. References

World Bank CCKP – Syrian Arab Republic. https://climateknowledgeportal.worldbank.org/

FAO. FAOSTAT – Crops and Livestock Products. https://www.fao.org/faostat/